A Minimal Holographic Model of Black Hole Singularity Resolution via Information Saturation

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We propose a minimal model in which the black hole singularity is replaced by an information-saturation threshold. As collapsing matter is compressed, its internal structure is progressively broken down into more elementary degrees of freedom, whose number is bounded above by the holographic limit $I_{\max} = A/4\ell_p^2$. Collapse terminates when this bound is saturated ($I = I_{\max}$); no further stable compression is possible, and the configuration undergoes a rupture that initiates a new expanding spacetime region. A modified Friedmann equation incorporating the saturation effect admits an exact bounce solution under radiation domination, requiring no exotic matter or higher-order curvature terms. We compute the resulting tensor-mode spectrum and find exponentially suppressed particle production, distinct from mechanisms driven by a finite-width confinement feature. The model requires no new fundamental fields and is presented as a conceptual and phenomenological proposal rather than a completed derivation.

I. INTRODUCTION

General relativity predicts the formation of singularities under realistic gravitational collapse [1, 2]: curvature and density diverge, and predictability is lost. Cosmic censorship hides these singularities behind horizons, but their existence inside black holes remains unresolved.

Loop quantum gravity [7], string theory, and holographic approaches [3, 4] have each made progress toward singularity resolution. The holographic principle in particular bounds the maximum information content of a spatial region by the area of its boundary, giving a natural ultraviolet cutoff on gravitational degrees of freedom.

We propose that the classical singularity is replaced by an information-saturation threshold. As the collapse radius decreases, matter is broken down into progressively more elementary degrees of freedom, without our needing to specify what those degrees of freedom ultimately are. Collapse halts when their number saturates the holographic bound, triggering a rupture and subsequent expansion.

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II. INFORMATION SATURATION

Let $I(r)$ denote the effective number of independent microscopic degrees of freedom compatible with the macroscopic configuration at radius r. This is consistent with coarse-grained entropy, entanglement entropy across local horizons, and Bekenstein–Hawking entropy on holographic screens. We remain agnostic about the underlying microphysics: we require only that $I(r)$ increase monotonically as r decreases, and that it be bounded above by the holographic limit

$$I_{\max}(r) = \frac{A(r)}{4\ell_p^2}, \qquad A(r) = 4\pi r^2. \tag{1}$$

The saturation radius r_0 is defined by

$$I(r_0) = I_{\max}(r_0). \tag{2}$$

As r decreases toward r_0, composite structure is progressively broken down and encoded on the boundary; the physical content of this process is that the number of independent bulk degrees of freedom is capped by the surface, not that matter passes through any particular sequence of lower-dimensional forms. At $r = r_0$, further compression would violate the bound. The configuration becomes unstable and undergoes a rupture, releasing its information into a new expanding spacetime region. This is not a classical topology change, which general relativity forbids, but a semiclassical branch transition of the kind expected in quantum gravity.

TABLE I. Comparison with selected singularity-resolution frameworks.

Approach	Singularity Resolved?	Mechanism	Holographic?
Loop Quantum Gravity	Yes	Quantum geometry repulsion	Partial
String Theory	Yes	Fuzzballs / Branes	Yes
Entropic Gravity	Partial	Thermodynamic force	Yes
This Model	**Proposed**	**Information saturation + rupture**	**Yes**

III. EFFECTIVE DYNAMICS AND BOUNCE

We model the saturation effect with a minimal modification to the Friedmann equation (natural units, $G = c = 1$, so ρ absorbs the usual $8\pi G/3$):

$$H^2 = \rho\left(1 - \frac{\rho}{\rho_{\mathrm{sat}}}\right), \tag{3}$$

where ρ_{sat} is the saturation density implied by the holographic bound. This is the lowest-order correction enforcing $H = 0$ at saturation, consistent with imposing a maximum entropy density in Jacobson's thermodynamic derivation of Einstein's equations [5]; the correction is thermodynamic, not material, and requires no violation of energy conditions.

As a heuristic illustration of how such a ceiling could arise from an underlying relational substrate, consider a toy mutual-information (MI) Laplacian model with partition function $Z =$

$\sum_n e^{-\beta\lambda_n}$. As the long-range correlation parameter $\alpha \to 0^+$, the underlying graph tends to the complete graph K_N by construction, giving a degenerate spectrum ($\lambda_n = N$) and an energy-density ceiling ($\rho_{\rm eff} = N$). This limit is a definitional feature of the toy ansatz, not an independently derived result, and we do not present it as a first-principles derivation of $\rho_{\rm sat}$. A non-tautological derivation of $\rho_{\rm sat}$ from microphysics remains open.

A. Exact bounce solution and tensor spectrum

For a radiation-dominated matter sector ($\rho \propto a^{-4}$), the modified Friedmann equation admits an exact closed-form solution,

$$a(t) = a_B \left[1 + (t/t_*)^2\right]^{1/4}, \qquad t_* = \frac{1}{2\sqrt{\rho_{\rm sat}}}, \tag{4}$$

with the same functional form as the standard loop-quantum-cosmology radiation bounce, though motivated here by information saturation rather than holonomy corrections. The conformal-time potential $U(\tau) = a''/a$ is likewise closed-form and smooth everywhere, peaking at the bounce.

We numerically integrated the tensor Mukhanov–Sasaki equation on this background with adiabatic vacuum initial conditions (Wronskian conserved to better than 10^{-7}) and extracted the Bogoliubov coefficient β_k. Unlike mechanisms with a finite-width confinement feature, which can produce a scale-free power-law tail in $|\beta_k|^2$, the smooth single-timescale bounce here gives exponentially suppressed particle production,

$$|\beta_k|^2 \sim e^{-2\kappa k}, \qquad \kappa\,\mathcal{H}_{\rm max} \approx 0.76, \tag{5}$$

where $\mathcal{H}_{\rm max}$ is the peak comoving Hubble rate at the bounce. The dimensionless product $\kappa\,\mathcal{H}_{\rm max}$ is constant across the values of $\rho_{\rm sat}$ tested, confirming a genuine kinematic scaling relation. This result assumes pure radiation domination and should not be conflated with the power-law tensor tilt found in models with an explicit confinement profile.

IV. RELATION TO EXISTING APPROACHES

This model is aligned with thermodynamic and holographic approaches to gravity. Its novelty lies in identifying information saturation, rather than exotic matter or metric quantization, as the direct trigger for the bounce.

Loop quantum gravity resolves the singularity via holonomy corrections and discrete geometry [7]; string-theoretic fuzzballs replace the interior with horizon-scale microstructure; entropic gravity treats gravity as emergent from entropy gradients [6]. The bounce equation derived here takes the same functional form as the LQC bounce under radiation domination, but is motivated by a different physical mechanism: information saturation on a relational substrate rather than quantum discreteness of geometry. We note this mathematical coincidence explicitly rather than treating the two models as independent.

V. DISCUSSION

A. Implications

Black Hole Interiors: The interior is a dynamic saturation-and-rupture region rather than a singularity; the surface at r_0 encodes the full information content of the collapsed matter.

Cosmology: Our universe may have originated from such a saturation event in a parent spacetime, offering a mechanism for cosmological bounces without exotic matter or fine-tuned scalar fields.

Information Paradox: Information is preserved through the saturation–rupture transition, with the holographic encoding at r_0 providing a natural bookkeeping mechanism.

B. Observational Signatures

- Gravitational wave echoes from near-horizon structure.
- Modifications to late-stage collapse dynamics.
- Deviations in Hawking radiation spectra due to saturation effects.
- Exponentially suppressed tensor particle production near the bounce (Sec. III.A), under the radiation-domination assumption. Whether this yields an observationally distinctive imprint, and how the result generalizes to other matter content, remains open.

C. Limitations and Future Work

This is a conceptual and phenomenological model. Equation (3) is posited on physical grounds rather than derived from a complete microphysical theory. We have computed the tensor spectrum under radiation domination (Sec. III.A), but the non-linear rupture dynamics have not been simulated, and the tensor calculation has not been extended beyond radiation domination. Future work includes a numerical treatment of the rupture phase, a non-tautological derivation of ρ_{sat}, generalization of the tensor calculation to other matter content, and comparison with gravitational-wave and cosmological observations.

VI. CONCLUSION

We have presented a minimal holographic model in which black hole singularities may be resolved by information saturation: collapse ends at a finite radius where the holographic bound is reached, triggering a rupture and the birth of a new expanding spacetime. The mechanism admits an exact bounce solution under radiation domination and a corresponding, distinctive prediction of exponentially suppressed tensor particle production. The model is offered as a theoretical and phenomenological proposal; its central claims await a non-tautological derivation of ρ_{sat} and numerical treatment of the rupture dynamics.

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